

YEAR 10 SCIENCE (GENERAL)
PHYSICAL SCIENCE
TEST 2

Name: SOLUTIONS

Mark: 52

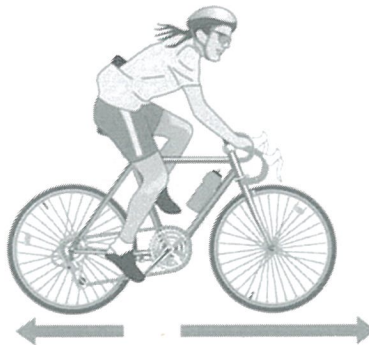
Part 1: Multiple Choice

Write your choice for each question in the table below.

1. The crumple zone of a car is designed to:
 - (a) allow the vehicle and passengers to stop more slowly.
 - (b) decrease the time before the car comes to a stop after a collision.
 - (c) allow the chassis to bend.
 - (d) create a reaction force that equals the action force.
2. A car manufacturer wants to produce cars that accelerate faster. Which changes in their cars would result in greater acceleration?
 - (a) Increase both the forward force by the engine and the mass of the car.
 - (b) Decrease both the forward force and the mass of the car.
 - (c) Increase the forward force and decrease the mass of the car.
 - (d) Decrease the forward force and increase the mass of the car.
3. The rate of change of speed of an object is known as:
 - (a) average speed.
 - (b) average velocity.
 - (c) change in velocity.
 - (d) acceleration.
4. An object starts from rest and accelerates at 5.0 m/s^2 . After 4.0 seconds, its velocity will be:
 - (a) 0.
 - (b) 10 m/s.
 - (c) 20 m/s.
 - (d) 30 m/s.
5. The tendency of an object to resist changes in its motion is called:
 - (a) inertia.
 - (b) weight.
 - (c) friction.
 - (d) work.
6. Newton's law, known as the law of inertia, is:
 - (a) his first law.
 - (b) his second law.
 - (c) his third law.
 - (d) none of the above.

MULTIPLE CHOICE ANSWERS	
1	A
2	C
3	D
4	C
5	A
6	A
7	B
8	C
9	B
10	B
11	A
12	C
13	A
14	A

7. An object is acted upon by a force of 50 N pushing it forwards and a total frictional force of 40 N acting in the opposite direction. The nett force will be:
- (a) 10 N backwards.
 - (b) 10 N forwards.
 - (c) 90 N backwards.
 - (d) 90 N forwards.
8. The size of the acceleration acting on an object of mass 10 kg that has a nett force of 3.0 N acting on it will be:
- (a) 13 m/s^2 .
 - (b) 3.33 m/s^2 .
 - (c) 0.3 m/s^2 .
 - (d) 30 m/s^2 .
9. The force of gravity acting on an object is called:
- (a) inertia.
 - (b) weight.
 - (c) friction.
 - (d) work.
10. The weight force acting on a 20 kg mass on the Earth would be:
- (a) 19.6 N.
 - (b) 196 N.
 - (c) 1.96 N.
 - (d) 0 N.
11. In the diagram below, the arrows show the size and direction of two force arrows acting on the bike.



Which of the following describes the effect of the pair of forces on the bike?

- (a) The bike will accelerate forward.
- (b) The bike will change its direction of motion.
- (c) The bike will remain stationary.
- (d) The bike will decelerate.

12. What is the acceleration of a car if it increases its velocity from 5.0 m/s to 15.0 m/s in 4.0 seconds?
- (a) 2.5 m/s
 - (b) 15 m/s²
 - (c) 2.5 m/s²
 - (d) 5.2 m/s²
13. A class is studying Newton's three laws. The teacher asks a student wearing rollerblades to face the wall and push against it. The student moves backwards away from the wall. Which is the best explanation for her movement?
- (a) Her action of pushing resulted in a reaction force by the wall, in the opposite direction to her push.
 - (b) She used her feet to push herself away from the wall.
 - (c) The push is a force and causes her to accelerate.
 - (d) The inertia of the student is responsible for her acceleration.
14. Which of the following statements is **correct**?
- (a) A car at rest has at least two forces acting on it.
 - (b) A car at rest has no forces acting on it.
 - (c) When you take your foot off the car's accelerator, the car slows down because there are no more forces acting on it.
 - (d) When you brake in a car that is rolling down a hill, the car stops because all of the forces add up to zero.

Part 2: Short Answer

Working to extended answer calculations should be set out clearly and logically.

1. Explain why it is dangerous to have large parcels or sharp objects loose in the back of a moving car. (2 marks)

- If the car stops suddenly, the objects can move forward and injure the passengers. (1)
- They have inertia and continue to move at the speed of the car. (1)

2. You have researched a safety feature for this topic.

(a) Name your safety feature.

- (b) Explain the main Physics principle behind how it acts to protect passengers. (3 marks)

SEAT BELT

- Newton's 1st law
- Mention inertia and moving forward.
- Force exerted on passengers to hold them in place.

AIR BAG / CRUMPLE ZONES

- Newton's 2nd law
- Mention $F \propto \frac{1}{t}$
- Increased time means smaller force experienced.

ABS

- Newton's 3rd law
- Action of brakes, reaction of car.

3. Describe the motion of a ball placed on the middle of the back seat of a car in the following situations.

- (a) The car brakes suddenly. Moves forward (1 mark)
- (b) The car turns right. Moves left (1 mark)
- (c) The car accelerates. Moves backwards (1 mark)

4. Complete the following table.

(3 marks)

Force	Mass	Acceleration
6000 N	1200 kg	5 m/s ²
150 N	25 kg	6.0 m/s ²
60 N	24 kg	2.5 m/s ²

[1 mark each]

5. A dragster accelerates from rest at 14.5 m/s² for 6.90 s during a race. What is the velocity of the vehicle after this time? (3 marks)

$$v = ?$$

$$u = 0 \text{ m/s}$$

$$a = 14.5 \text{ m/s}^2$$

$$t = 6.90 \text{ s}$$

Forwards +ve

$$v = u + at \quad (1)$$

$$= 0 + (14.5)(6.90) \quad (1)$$

$$= \underline{100 \text{ m/s forwards}} \quad (1)$$

6. How long does it take for a 1400 kg Supercar to reach 190 km/h if it accelerates from the start line in a race at 12.6 m/s²? (3 marks)

Forwards +ve

$$v = 52.8 \text{ m/s}$$

$$u = 0 \text{ m/s}$$

$$a = 12.6 \text{ m/s}^2$$

$$t = ?$$

$$a = \frac{v-u}{t}$$

$$\Rightarrow t = \frac{v-u}{a} \quad (1)$$

$$= \frac{52.8 - 0}{12.6} \quad (1)$$

$$= \underline{4.19 \text{ s}} \quad (1)$$

7. A parachutist is 1000 m above the ground when she pops her chute. What is her velocity at this point if the parachute decelerates her at 27.0 m/s^2 and drops her velocity to 80.0 km/h in 1.10 s ? (3 marks)

Down is +ve

$$v = 22.2 \text{ m/s}$$

$$u = ?$$

$$a = -27.0 \text{ m/s}^2$$

$$t = 1.10 \text{ s}$$

$$v = u + at$$

$$\Rightarrow u = v - at \quad (1)$$

$$= 22.2 - (-27.0)(1.10) \quad (1)$$

$$= \underline{51.9 \text{ m/s down}} \quad (1)$$

8. A Space Shuttle lands at 270 km/h and immediately pops a drag chute to assist its brakes to stop the vehicle. Assume that the braking forces are uniform for this motion.



- (a) If it takes 15.0 s to stop, calculate the **deceleration** experienced by the crew. (3 marks)

Forward is +ve

$$v = 0 \text{ m/s}$$

$$u = 75.0 \text{ m/s}$$

$$a = ?$$

$$t = 15.0$$

$$a = \frac{v - u}{t} \quad (1)$$

$$= \frac{0 - 75.0}{15.0} \quad (1)$$

$$= \underline{-5.00 \text{ m/s}^2 \text{ backwards}} \quad (1)$$

- (b) If the mass of the Shuttle is $34,500 \text{ kg}$, what is the average force exerted by the brakes and the chute?

(If you could not find an answer for part (a), use deceleration = -6.0 m/s^2 .)

(3 marks)

$$F = ?$$

$$m = 34,500 \text{ kg}$$

$$a = -5.00 \text{ m/s}^2$$

Forwards is +ve

$$F = ma \quad (1)$$

$$= (34,500)(-5.00) \quad (1)$$

$$= \underline{-172,500 \text{ N backwards}} \quad (1)$$

9. A car of mass 1750 kg is accelerated from 70.0 km/h by the engine supplying a force of 2750 N. The frictional forces acting on the car total 375 N.

(a) What is the nett force acting on the car?

(2 marks)

Forwards is +ve

$$\Sigma F = 2750 - 375 \quad (1)$$
$$= \underline{2375 \text{ N forwards}} \quad (1)$$

(b) What is the velocity of the car after 11.0 s of acceleration?

(5 marks)

$$\Sigma F = 2375 \text{ N}$$
$$m = 1750 \text{ kg}$$
$$a = ?$$

Forwards is +ve

$$\Sigma F = ma$$
$$\Rightarrow a = \frac{\Sigma F}{m} \quad (1)$$
$$= \frac{2375}{1750} \quad (1)$$
$$= 1.36 \text{ m/s}^2 \text{ forwards} \quad (1)$$

$$v = ?$$
$$u = 19.4 \text{ m/s}$$
$$a = 1.36 \text{ m/s}^2$$
$$t = 11.0 \text{ s}$$

$$v = u + at$$
$$= 19.4 + (1.36)(11.0) \quad (1)$$
$$= \underline{34.4 \text{ m/s forwards}} \quad (1)$$

10. An object has a mass of 6.0 kg.

(a) What is its **weight** on Earth?

(2 marks)

$$F_w = ?$$

$$m = 6.0 \text{ kg}$$

$$g = 9.80 \text{ m/s}^2$$

$$F_w = mg$$

$$= (6.0)(9.80) \quad (1)$$

$$= \underline{58.8 \text{ N}} \quad (1)$$

(b) Would the object **weigh more or less** on the Moon? Explain your answer.

(2 marks)

• less. (1)

• less gravity so g is less. (1)

(c) What is its **mass** on the Moon?

(1 mark)

$$6.0 \text{ kg} \quad (1)$$

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PHYSICAL SCIENCES
DATA SHEET

Speed $\text{speed} = \frac{d}{t}$

Velocity $v = \frac{s}{t}$

Acceleration $a = \frac{v - u}{t}$

$$v = u + at$$

Force $F = ma$

Weight $F_w = mg$

Acceleration due to gravity $g = 9.80 \text{ ms}^{-2}$

Converting kmh^{-1} to ms^{-1} Divide the speed in kmh^{-1} by 3.6 to get the speed in ms^{-1} .